

KIRAN BHATT

Kathmandu, Nepal | bhattkiran735@gmail.com

Biotechnology graduate working across wet lab microbiology, molecular diagnostics, and data engineering, with research spanning antimicrobial resistance, genomics, and applied data pipelines.

EDUCATION

BSc (Hons) Biotechnology | BRAC University, Dhaka, Bangladesh

March 2025

GPA: 3.91/4.0 (Summa Cum Laude equivalent); BRAC University Merit Scholarship (fully funded, held throughout the degree); Stanford Data Ocean Course Scholarship

RESEARCH

Undergraduate Thesis | BRAC University, Department of Biotechnology

Nov 2024

Multidrug Resistant Pathogens in Biomedical Waste from Tertiary Hospitals in Bangladesh: Molecular Identification and Antimicrobial Resistance Profiles

- Isolated and molecularly confirmed 87 bacterial isolates (*Pseudomonas*, *E. coli*, *Shigella*, and *Salmonella* spp.) from biomedical waste collected across five tertiary hospitals, using species-specific PCR (16S rRNA, *invA*, *ipaH*, and *uidA*-targeted primers) following biochemical presumptive identification
- Ran antibiotic susceptibility testing by Kirby-Bauer disc diffusion (CLSI guidelines) across a panel of 21 antibiotics; found ampicillin resistance up to 100% in *Salmonella* and *Shigella* spp. and multidrug resistance prevalence up to 91.7% (*Salmonella* spp.).
- Contributed to data curation and visualization of resistance profiles, including heatmaps, hierarchical clustering dendrograms, and PCA plots generated in R

DATA ENGINEERING & COMPUTATIONAL WORK

Data Engineering Trainee | Data Engineering Training Program

June-July 2026

- Built a METABRIC clinical data engineering platform as a capstone project: Bronze/Silver/Gold layered pipeline with Airflow orchestration and a Streamlit dashboard
- Built ETL pipelines and practiced database design (star schema, OLTP/OLAP) using SQL, Python, and Apache Airflow on real-world datasets, including ride-sharing trip data

Applied Data Institute Scholar | Equitech Futures

Sep – Dec 2025

Global Research Scientist Cohort (competitively selected, interdisciplinary)

- Partnered with Athena Infonomics on a water and sanitation utility performance dashboard; converted stakeholder notes into a structured user profile before any wireframing began
- Built the metrics and reporting layer across financial, operational, and service-coverage indicators, from wireframe through final dashboard

Precision Medicine & Genomics Coursework | Stanford Data Ocean

Jan – Dec 2025

- Queried population-scale genomic databases and the COSMIC cancer mutation database using Python and SQL in a cloud environment
- Applied ROC curve analysis to compare predictive performance across multi-omic data layers for infection-state classification
- Applied supervised and unsupervised learning methods to wearable-device and clinical datasets for outcome classification

LABORATORY EXPERIENCE

- Research Intern | Bangladesh Council of Scientific and Industrial Research (BCSIR),Dhaka May 2024
- Verified species composition in food products using PCR-based molecular diagnostics; tested therapeutic drug levels in animal samples using standard analytical protocols
 - Performed microbiological quality control testing, including microbial culture, isolation, and biochemical identification
 - Maintained laboratory records to GLP documentation standards

TECHNICAL SKILLS

Data Engineering: SQL, Python (Pandas, NumPy), ETL pipeline design, Apache Airflow, data warehousing, database modeling (OLTP/OLAP, star schema), DBeaver, psycopg, Streamlit

Data Science & Statistics: R (heatmaps, hierarchical clustering, PCA), ROC curve analysis, biostatistics, multi-omic data integration, population-scale variant analysis (gnomAD, ClinVar, COSMIC), supervised and unsupervised learning

Molecular & Cell Biology: PCR (standard and quantitative, species-specific primer design), DNA extraction, gel electrophoresis, SDS-PAGE, Western blotting, ELISA, brightfield microscopy

Microbiology & Quality Control: microbial culture, isolation, and biochemical identification; antibiotic susceptibility testing (Kirby-Bauer disc diffusion, CLSI guidelines); aseptic technique; GLP documentation and quality assurance

LANGUAGES

Fluent in English, Nepali, Hindi, and Bangla